DISCRETE RANDOM VARIABLE

1. A random variable X has the following probability distribution

x	0	1	2	3	4	5	6	7
$P(x)$	0	k	$2k$	$2k$	$3k$	k^2	$2k^2$	$7k^2+k$

Find $i)\ k$ $ii)\ P(x<6)$ $iii)\ P(x\geq 6)$ $iv)\ P(0<x<5)$.

2. The outcomes of a certain experiment are $x_1=1, x_2=2, x_3=3$. The associated probability function is $P(x)=\begin{cases} k & x=1 \\ 2k & x=2 \\ 5k & x=3 \end{cases}$ Find $P(x<2), P(x\leq 2), P(x\leq 3)$.

3. Given the following probability function,

x	0	1	2	3	4	5
$P(x)$	0	c	2c	2c	c	$7c^2$

Find $i)\ c$ $ii)\ P(x\geq 2)$ $iii)\ P(x<3)$.

4. A discrete random variable X has a probability mass function as $P(x)=cx$ for $x=0,1,2,3,4,5$. Find (i) c (ii) $P(X\leq 4)$ (iii) Mean of X.

5. Let X be a discrete random variable with the following PMF $P(x)=\begin{cases} 0.1 & for\ x=0.2 \\ 0.2 & for\ x=0.4 \\ 0.2 & for\ x=0.5 \\ 0.3 & for\ x=0.8 \\ k & for\ x=1 \\ 0 & otherwise \end{cases}$

   a. Find R_X, the range of the random variable X.
   b. Find k.
   c. Find $P(X\leq 0.5)$.
   d. Find $P(0.25<X<0.75)$.

MATHEMATICAL EXPECTATION, MEAN AND VARIANCE

1. A die is tossed once. Random variable X, denote the digit that appears. Find the Mean (expected value) and Variance.

2. A die is tossed twice. X denotes the sum of digits in two throws. Find the Mean (expected value) and Variance.
3. A coin is thrown 4 times successively. X denotes the number of heads. Find the Mean and Variance.
4. Variable x takes the values $0, 1, 2, 3, 4, 5$ with probability of each as $\frac{1}{15}, \frac{2}{15}, \frac{3}{15}, \frac{3}{15}, \frac{5}{15}, \frac{1}{15}$. Find the Mean and Variance.
5. the probabilities that a customer will busy 1,2,3,4,5 items in a grocery store are $\frac{3}{10}, \frac{1}{10}, \frac{1}{10}, \frac{2}{10}, \frac{3}{10}$ respectively. What is the average number of items that a customer will busy?

BINOMIAL PROBABILITY DISTRIBUTIONS

1. Derive the mean and variance of the binomial distribution.
2. An unbiased coin is tossed 10 times. Find the probability of getting exactly 6 heads, at least 6 heads.
3. Probability of Man aged 60 years will live for 70 years is 1/10. Find the probability of 5 men selected at random 2 will live for 70 years.
4. 20% bolts are produced by machine are defective. Find the probability that out of three bolts are chosen at random 1 is defective.
5. The probability that hit a target in shooting practice is 0.3. If the shoots 10 times, find the probability that he hits the target.
6. A box contains 100 bulbs out of which 10 are defective. A sample of 5 bulbs is drawn. Find the probability that none is defective.
7. Out of 2000 families with 4 children each, find the number of families you would expect to have (i) at least one boy. (ii) two boys. (iii) no girl.
8. The mean and variance of Binomial probability distribution are 6 and 2 resp. Find $p(r \geqslant 2)$.

9. An insurance agent accepts policies of 5 men of identical age and in good health. The probability that a man of this age will live 30 years hence is 2/3. Find the probability that in 30 years (i) all five men and (ii) at least one man will be alive.
10. On an average, a box containing 10 articles is likely to have 2 defectives. If we consider a consignment of 100 boxes,
    (i) how many of them are expected to have at the most one defective?
    (ii) how many of them are expected to have three or less defective?
11. On an average 20% of the workers in an industry suffer with a certain disease. If 12 workers are chosen from the industry, find the probability that (i) Exactly 2 workers suffer from the disease, (ii) At least one worker suffers from the disease.
12. A coffee connoisseur claims that he can distinguish between a cup of instant coffee and a cup percolator coffee 75% of the time. It is agreed that his claim will be accepted if he correctly identifies at least 5 of the 6 cups. Find the chances of having the claim (i) accepted (ii) rejected, when he does have the ability of claims.
13. A series of five one day matches is to be played between India and Pakistan. Assuming that the probability of India's win in each match as 0.6 and results of all the five matches independent of each other, find the probability that (i) India wins at least one match, (ii) India wins the series.
14. In a binomial probability distribution consisting of 5 independent trials, probabilities of 1 and 2 successes are found to be 0.4096 and 0.2048 respectively. Find the parameter p of the distribution.
15. With the usual notations, find the probability of success p for a binomial distribution, if $n = 6$, and $9P(x = 4) = P(x = 2)$.

POISSON PROBABILITY DISTRIBUTION

1. Derive the mean and variance of the Poisson distribution.
2. In a certain factory turning out razor blades, there is a small chance of 1/500 for any blade to be defective. The blades are supplied in a packet of 10. Use Poisson distribution

to calculate number of packets containing no defective and two defective blades, in a consignment of 10000 packets.

3. The average number of misprints per page of a book is 1.5. Assuming the distribution of number of misprints to be Poisson. Find (i) the prob that a particular book is free from misprints. (ii) Number of pages containing more than one misprint if the book contains 900 pages.
4. Number of road accidents on a highway during a month follows a Poisson distribution with mean 5. Find the probability that in a certain month number of accidents on the highway will be (i) less than 3. (ii) Between 3 & 5. (iii) more than 3.
5. A manufacturer of cotter pins knows that 2% of his product is defective. If he sells cotter pins in boxes of 100 pins and guarantees that not more than 5 pins will be defective in a box, find the app probability that a box will fail to meet the guaranteed quality.
6. Between 2 pm and 3 pm, the average numbers of phone calls per minute coming into the company are 2. Find the probability that during one particular minute, there will be [i] no phone call at all. [ii] 2 or less calls.
7. If the probability that a concrete cube fails is 0.001. Determine the probability that out of 1000 cubes: [i] Exactly two [ii] More than one cube will fail.
8. There is a small probability of 1/1,000 for any computer produced to be defective. Determine in a sample of 2,000 computers, the probability that there (i) is no defective and (ii) are two defectives.